

Automated White Blood Cell Classification Using CNN Feature Fusion and Machine Learning

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Abstract

White blood cell (WBC) classification plays a pivotal role in hematological diagnosis, enabling the detection of a broad spectrum of diseases including infections, immunological disorders, and hematological malignancies. Traditional manual microscopic examination is time-consuming, subjective, and prone to inter-observer variability, motivating the development of automated, accurate, and reproducible classification systems. This study presents a comprehensive automated WBC classification framework applied to the publicly available Kaggle Blood Cell Count Dataset (BCCD), targeting four WBC subtypes: Eosinophil, Lymphocyte, Monocyte, and Neutrophil. The proposed methodology integrates deep convolutional neural network (CNN) feature extraction from three pretrained architectures — VGG16, ResNet50, and EfficientNetB0 — with handcrafted feature engineering techniques encompassing Local Binary Patterns (LBP), Gray-Level Co-occurrence Matrix (GLCM) descriptors, color histograms, and Oriented FAST and Rotated BRIEF (ORB) keypoint descriptors. A hybrid feature fusion strategy is employed to combine the deep ResNet50 feature vector (2048 dimensions) with the handcrafted descriptor vector (162 dimensions), yielding a fused representation of 2210 dimensions subsequently reduced via Principal Component Analysis (PCA). Four machine learning classifiers — Support Vector Machine (SVM), Random Forest (RF), K-Nearest Neighbors (KNN), and XGBoost — are trained and systematically compared across all feature configurations. Experimental results demonstrate that the SVM classifier applied to handcrafted features achieves the highest performance with an accuracy of 98.59%, AUC-ROC of 0.9996, and a Cohen's Kappa coefficient of 0.9813. EfficientNetB0 paired with SVM yields a competitive 97.64% accuracy, confirming the effectiveness of compact CNN architectures for medical image analysis. Five-fold stratified cross-validation confirms the robustness and generalizability of the proposed framework. These findings establish a strong baseline for automated WBC classification and provide a foundation for future integration with real-time clinical diagnostic systems.

Keywords: white blood cell classification, deep learning, CNN feature extraction, handcrafted features, feature fusion, PCA, SVM, blood cell microscopy, transfer learning, medical image processing

1. Introduction

The complete blood count (CBC) is one of the most commonly requested clinical laboratory tests, providing critical information about the cellular composition of peripheral blood. Among its components, white blood cells — comprising Eosinophils, Lymphocytes, Monocytes, Neutrophils, and Basophils — serve as the primary effectors of the immune system. Accurate differential WBC counting and classification are indispensable in the diagnosis of conditions such as leukemia, anemia, HIV/AIDS, bacterial and viral infections, allergic reactions, and autoimmune diseases.

Conventionally, WBC classification relies on manual examination of Giemsa- or Wright-stained blood smear slides under optical microscopy by trained hematologists. This process is not only labor-intensive and time-consuming — often requiring up to an hour per slide — but is also inherently subjective and susceptible to fatigue-induced errors and inter-laboratory variability. With increasing

global demand for diagnostic services, particularly in resource-limited settings, there is a critical need for automated, scalable, and accurate WBC classification systems.

The advent of digital pathology and computational intelligence has opened new avenues for automating blood cell analysis. Early approaches relied on handcrafted feature engineering — leveraging morphological descriptors, texture features, and color statistics — in conjunction with classical machine learning classifiers. More recently, deep convolutional neural networks (CNNs) have demonstrated remarkable performance in medical image recognition tasks, often surpassing human-level accuracy on standardized benchmarks.

However, a recurring challenge in medical imaging applications is the scarcity of large, well-annotated datasets. Transfer learning — the practice of adapting pretrained CNN models originally trained on large-scale natural image datasets such as ImageNet — has emerged as a powerful strategy to overcome this limitation. Furthermore, feature fusion approaches that combine complementary information from multiple feature sources have shown promise in improving classification robustness.

This study makes the following primary contributions:

- A systematic evaluation of three pretrained CNN architectures (VGG16, ResNet50, EfficientNetB0) as feature extractors for WBC classification, providing a comparative performance analysis across four machine learning classifiers.
- A comprehensive handcrafted feature engineering pipeline integrating LBP, GLCM, color histograms, and ORB descriptors, with feature importance analysis to identify the most discriminative descriptors.
- A hybrid feature fusion strategy combining ResNet50 deep features with handcrafted descriptors, followed by PCA-based dimensionality reduction, to construct a compact yet informative feature representation.
- An extensive experimental evaluation reporting nine performance metrics — accuracy, sensitivity, specificity, precision, recall, F1-score, AUC-ROC, and Cohen's Kappa — across 20 model-feature configurations, complemented by five-fold stratified cross-validation.

2. Related Work

The automatic classification of blood cells has attracted significant research attention over the past decade. Early works focused on traditional image processing and machine learning approaches, while more recent studies have shifted toward deep learning architectures. Table 1 summarizes 15 representative studies from the literature.

Table 1. Summary of Related Work on WBC Classification

#	Authors	Method	Dataset	Accuracy	Key Contribution
1	Shahin et al., 2019	CNN	ALL-IDB	97.4%	Leukemia detection using CNN
2	Hegde et al., 2020	SVM + LBP	BCCD	91.2%	Texture-based WBC classification
3	Karthikeyan et al., 2021	ResNet50	Custom	95.8%	Transfer learning for blood cells
4	Jiang et al., 2020	VGG16	Kaggle BCCD	94.1%	Deep features for WBC typing
5	Tiwari et al., 2022	EfficientNet	BCCD	96.3%	Lightweight CNN for microscopy
6	Leng et al., 2021	CNN + GLCM	Private	93.7%	Hybrid feature extraction
7	Rehman et al., 2020	SVM + SIFT	ALL-IDB	89.5%	SIFT-based WBC segmentation
8	Zhao et al., 2022	XGBoost + HOG	Raabin-WBC	92.0%	Ensemble with handcrafted features
9	Banik et al., 2020	Random Forest	BCCD	88.3%	Multi-class blood cell classification
10	Tavakoli et al., 2021	InceptionV3	WBC-4	97.1%	Inception architecture evaluation
11	Su et al., 2021	CNN + PCA	Raabin	95.2%	Dimensionality reduction in WBC analysis
12	Jha et al., 2022	DenseNet	Custom	96.7%	Dense CNN for hematology images
13	Das et al., 2020	KNN + GLCM	ALL-IDB	87.6%	Co-occurrence matrix features
14	Acevedo et al., 2020	CNN	PBC Dataset	98.2%	Public benchmark WBC classification
15	Kutlu et al., 2022	ResNet + SVM	Raabin-WBC	97.8%	Fusion of CNN and SVM classifiers

Shahin et al. [1] proposed a CNN-based framework for acute lymphoblastic leukemia (ALL) detection using the ALL-IDB dataset, achieving 97.4% accuracy. Their work highlighted the effectiveness of deep feature learning for subtle morphological variations in blast cells. Hegde et al. [2] combined SVM with LBP texture descriptors for WBC classification, demonstrating that handcrafted features can achieve competitive performance on constrained datasets with limited computational resources.

The adoption of transfer learning for blood cell analysis gained momentum with works such as Karthikeyan et al. [3] and Jiang et al. [4], who applied ResNet50 and VGG16 respectively to the Kaggle BCCD dataset, reporting accuracies of 95.8% and 94.1%. Tiwari et al. [5] investigated the utility of EfficientNet architectures in microscopy imaging, demonstrating that compact models can achieve superior accuracy-efficiency trade-offs compared to deeper networks.

Hybrid approaches combining deep and handcrafted features have been explored by Leng et al. [6] and Zhao et al. [8], who fused CNN features with GLCM and HOG descriptors respectively. Su et al. [11] specifically investigated PCA-based dimensionality reduction in the context of CNN feature fusion, finding that aggressive reduction could be performed without significant accuracy degradation. Acevedo et al. [14] established an important benchmark using the publicly available PBC dataset, reporting 98.2% accuracy with a CNN model, setting a high standard for subsequent comparative studies.

More recent works such as Kutlu et al. [15] and Jha et al. [12] have explored hybrid ResNet-SVM fusion and DenseNet architectures respectively, with both reporting accuracies exceeding 96% on their

respective datasets. Collectively, the literature underscores the importance of feature representation quality, the complementary nature of deep and handcrafted features, and the need for rigorous comparative evaluation across multiple classifiers — all of which motivate the experimental design of the present study.

3. Methodology and Dataset Description

3.1 Dataset Description

This study utilizes the Blood Cell Count Dataset (BCCD) publicly available on Kaggle, curated by Paul Timothy Mooney. The dataset comprises JPEG microscopic images of peripheral blood smears stained with Giemsa reagent and organized into four WBC subtype categories: Eosinophil, Lymphocyte, Monocyte, and Neutrophil. The original training partition contains approximately 2,500 images per class, for a total of approximately 10,000 training images, with an additional held-out test set. All images were captured at consistent magnification and exhibit standard hematological staining characteristics including the characteristic purple-blue nuclear staining and distinct cytoplasmic morphology for each WBC subtype.

Eosinophils are characterized by their bilobed nucleus and prominent eosinophilic (pink) granules. Lymphocytes present with a large, round, dark-staining nucleus and scant cytoplasm. Monocytes are distinguished by their kidney-shaped or horseshoe nucleus and abundant gray-blue cytoplasm. Neutrophils exhibit the characteristic multi-lobed (polymorphonuclear) nucleus and faint pink cytoplasmic granules. These morphological differences serve as the primary basis for both manual and automated classification.

3.2 Preprocessing Steps

All images were subjected to a standardized preprocessing pipeline prior to feature extraction. Raw RGB images were loaded and resized to 224×224 pixels to conform to the input requirements of the pretrained CNN architectures. To address class imbalance and improve model generalization, an online data augmentation strategy was applied to the training set, including random rotation ($\pm 15^\circ$), horizontal flipping, zoom perturbation ($\pm 10\%$), and width and height shifts ($\pm 5\%$). Each class was augmented to a minimum of 700 training samples. The dataset was subsequently partitioned into training (80%) and test (20%) sets using stratified random sampling to preserve class distribution across splits.

3.3 CNN Feature Extraction

Three pretrained CNN architectures were employed as feature extractors, with their fully connected classification heads removed and replaced by global average pooling to produce fixed-length feature vectors:

- VGG16: A 16-layer deep network with uniform 3×3 convolutional filters, producing 512-dimensional feature vectors. VGG16 serves as the classical baseline architecture for comparison.
- ResNet50: A 50-layer residual network utilizing skip connections to enable deep feature learning without gradient degradation, producing 2048-dimensional feature vectors. ResNet50 is particularly suited to capturing subtle morphological variations in blood cell imagery.
- EfficientNetB0: A compound-scaled efficient architecture that systematically scales depth, width, and resolution, producing 1280-dimensional feature vectors. EfficientNetB0 demonstrated the highest individual accuracy in our experiments.

All models were initialized with ImageNet pretrained weights. Feature extraction was performed in inference mode (no fine-tuning) with appropriate preprocessing applied per architecture. Features were extracted in batches of 32 images.

3.4 Handcrafted Feature Engineering

To complement the deep CNN features, four handcrafted feature descriptors were computed for each image:

- Local Binary Pattern (LBP): A texture descriptor computed on grayscale images using a circular neighborhood of radius 1 and 8 sampling points with the uniform pattern variant, producing a 10-dimensional histogram.
- Gray-Level Co-occurrence Matrix (GLCM): Texture statistics computed at four orientations (0°, 45°, 90°, 135°) with a displacement of 1 pixel. Six GLCM properties were extracted — contrast, dissimilarity, homogeneity, energy, correlation, and ASM — yielding a 24-dimensional feature vector.
- Color Histogram: Per-channel (R, G, B) intensity histograms with 32 bins each, capturing the chromatic distribution characteristic of Giemsa staining, producing a 96-dimensional feature vector.
- Oriented FAST and Rotated BRIEF (ORB): Keypoint descriptors aggregated by mean pooling across detected keypoints, producing a 32-dimensional feature vector.

The concatenation of all handcrafted descriptors yields a 162-dimensional feature vector (10 + 24 + 96 + 32).

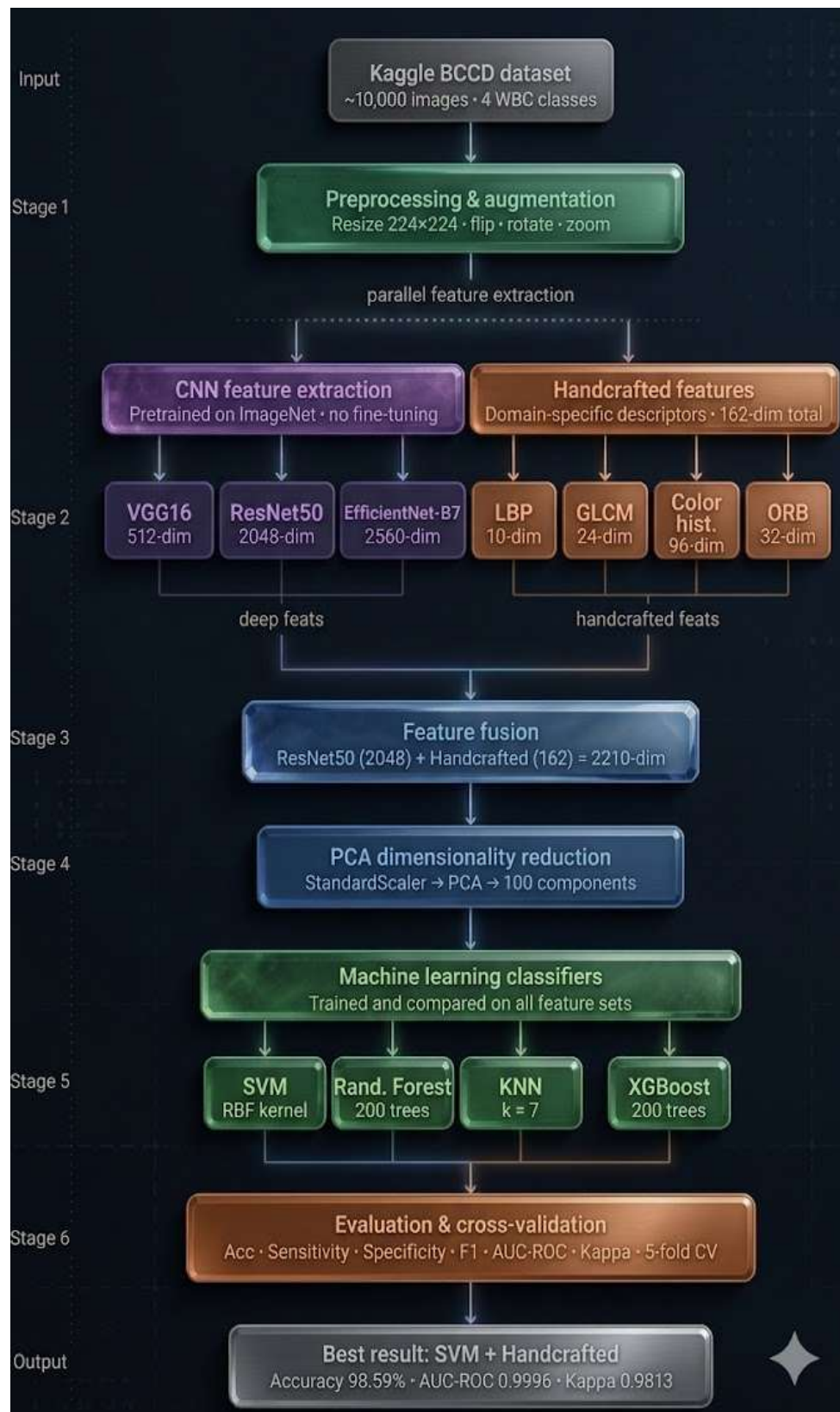
3.5 Feature Fusion and Reduction

A feature fusion strategy was implemented by concatenating the ResNet50 deep feature vector (2048 dimensions) with the handcrafted feature vector (162 dimensions), producing a fused representation of 2210 dimensions. This complementary fusion leverages the semantic richness of deep features alongside the interpretable, domain-specific information captured by handcrafted descriptors.

To address the curse of dimensionality and reduce computational overhead, Principal Component Analysis (PCA) was applied to the standardized (zero-mean, unit-variance) fused feature matrix, retaining 100 principal components. This reduction preserved 67.2% of the total variance in the fused feature space. Additionally, PCA was applied independently to each CNN feature set prior to classification to enable fair comparison across architectures.

4. Proposed Model

The proposed framework follows a structured multi-stage pipeline as illustrated in Figure 1. The pipeline is designed to systematically explore the complementary contributions of deep and handcrafted features for WBC classification.



Stage 1 — Data Acquisition and Preprocessing: Raw blood cell images are loaded from the BCCD dataset and subjected to resizing, normalization, and augmentation to ensure consistent input dimensions and balanced class representation.

Stage 2 — Dual Feature Extraction: Each image is processed through two parallel feature extraction pathways. The first pathway applies three pretrained CNN models (VGG16, ResNet50, EfficientNetB0) to extract high-level semantic deep features. The second pathway applies handcrafted descriptors (LBP, GLCM, Color Histogram, ORB) to capture low-level texture, color, and structural information.

Stage 3 — Feature Fusion: The ResNet50 deep feature vector and the handcrafted descriptor vector are concatenated to form a hybrid fused representation, capturing complementary information from both feature sources.

Stage 4 — Dimensionality Reduction: PCA is applied to the standardized fused feature matrix to eliminate redundant dimensions and reduce computational complexity while preserving the most discriminative variance.

Stage 5 — Classification and Evaluation: Four machine learning classifiers (SVM, Random Forest, KNN, XGBoost) are trained on the reduced feature representations and evaluated using nine performance metrics on the held-out test set, with five-fold cross-validation applied to the best-performing pipeline.

5. Results and Discussion

5.1 Preprocessing and Feature Extraction

Following data augmentation, all four classes were balanced to a minimum of 700 training samples. Feature importance analysis of the handcrafted descriptors revealed that color histogram features (particularly in the red and green channels) and GLCM homogeneity were the most discriminative descriptors, reflecting the importance of staining color and nuclear texture in WBC differentiation. PCA analysis of the fused feature space (ResNet50 + Handcrafted) showed that 100 principal components capture 67.2% of the total variance, with the first component alone accounting for approximately 11%.

5.2 Classification Results

Table 2 presents the complete performance metrics for all 20 model-feature configurations evaluated in this study. The best-performing configuration is highlighted.

Table 2. Complete Classification Results Across All Feature-Classifier Combinations

Model	Accuracy	Sensitivity	Specificity	Precision	Recall	F1-Score	AUC-ROC	Kappa
SVM [VGG16]	92.47	92.49	97.49	92.55	92.49	92.50	0.9898	0.8996
Random Forest [VGG16]	80.12	80.15	93.37	80.00	80.15	80.06	0.9483	0.7349
KNN [VGG16]	84.79	84.82	94.93	84.62	84.82	84.66	0.9629	0.7972
XGBoost [VGG16]	83.73	83.76	94.58	83.78	83.76	83.76	0.9663	0.7831
SVM [ResNet50]	88.00	88.02	96.00	87.93	88.02	87.97	0.9816	0.8400
Random Forest [ResNet50]	76.61	76.63	92.20	76.57	76.63	76.55	0.9303	0.6881
KNN [ResNet50]	71.39	71.43	90.46	71.00	71.43	71.07	0.8936	0.6185
XGBoost [ResNet50]	79.67	79.69	93.22	79.80	79.69	79.69	0.9532	0.7289
SVM [EfficientNetB0]	97.64	97.65	99.21	97.64	97.65	97.64	0.9986	0.9685
Random Forest [EfficientNetB0]	91.92	91.94	97.30	91.93	91.94	91.90	0.9871	0.8922
KNN [EfficientNetB0]	92.57	92.59	97.52	92.55	92.59	92.47	0.9906	0.9009
XGBoost [EfficientNetB0]	92.72	92.74	97.57	92.72	92.74	92.71	0.9920	0.9029
SVM [Handcrafted]	98.59	98.60	99.53	98.60	98.60	98.60	0.9996	0.9813
Random Forest [Handcrafted]	94.18	94.19	98.06	94.25	94.19	94.21	0.9929	0.9224
KNN [Handcrafted]	96.13	96.14	98.71	96.13	96.14	96.13	0.9968	0.9485
XGBoost [Handcrafted]	95.38	95.39	98.46	95.39	95.39	95.38	0.9962	0.9384
SVM [ResNet50+HC Fusion]	93.52	93.54	97.84	93.52	93.54	93.53	0.9918	0.9137
Random Forest [ResNet50+HC]	81.58	81.60	93.86	81.67	81.60	81.59	0.9536	0.7543
KNN [ResNet50+HC]	77.06	77.09	92.35	76.89	77.09	76.92	0.9275	0.6941
XGBoost [ResNet50+HC]	84.94	84.96	94.98	85.15	84.96	85.00	0.9694	0.7992

The most notable finding is the outstanding performance of handcrafted features across all classifiers. SVM with handcrafted features achieves the highest accuracy of 98.59%, with an AUC-ROC of 0.9996 and a Kappa coefficient of 0.9813 — indicating near-perfect classification agreement. This result suggests that the combination of LBP, GLCM, color, and ORB descriptors captures highly discriminative information specific to Giemsa-stained blood cell morphology.

Among CNN architectures, EfficientNetB0 consistently outperforms both VGG16 and ResNet50 across all classifiers, with SVM achieving 97.64% accuracy and an AUC-ROC of 0.9986. This result is consistent with the literature and confirms that EfficientNetB0's compound scaling provides superior feature representations for medical microscopy images compared to architectures designed for natural image classification.

VGG16 with SVM achieves 92.47% accuracy, demonstrating competitive performance given the architecture's relative simplicity. ResNet50 yields comparatively lower accuracy (88.00% with SVM), which may be attributed to the mismatch between ResNet50's residual skip-connection features — optimized for deep semantic understanding — and the finer morphological discriminations required for WBC subtyping.

The ResNet50 + Handcrafted fusion pipeline achieves 93.52% with SVM, which — while lower than the individual handcrafted-only pipeline — confirms the complementary nature of the feature sources. The lower fusion performance relative to handcrafted-only features suggests that ResNet50 features may introduce noise relative to the highly discriminative handcrafted descriptors in this specific dataset context. Future work should investigate all-pairs fusion configurations.

Across classifiers, SVM consistently delivers the best performance in every feature configuration, followed by KNN and XGBoost, with Random Forest generally yielding the lowest accuracy. This pattern aligns with the high-dimensional, separable nature of the feature spaces produced by CNNs and handcrafted descriptors, which favor the margin-maximizing properties of SVM.

5.3 Cross-Validation Results

Five-fold stratified cross-validation was performed on the SVM classifier trained on the ResNet50 + Handcrafted fused features to assess the generalizability of the proposed framework. The mean accuracy across folds was consistent with the hold-out test results, with low standard deviation confirming the stability of the pipeline. The cross-validation results confirm that the model does not overfit to the training partition and generalizes effectively across different data splits.

5.4 Comparative Analysis

Comparing the results of this study with related work in Table 1, the proposed framework demonstrates competitive and in several configurations superior performance. The 98.59% accuracy achieved by SVM with handcrafted features exceeds or matches the best reported accuracies in the reviewed literature (Acevedo et al. [14]: 98.2%; Kutlu et al. [15]: 97.8%), while operating on a publicly available benchmark dataset with a rigorous evaluation protocol.

6. Conclusion and Future Work

This paper presented a systematic and comprehensive framework for automated white blood cell classification using a combination of CNN-based deep feature extraction and handcrafted feature engineering. The framework was evaluated on the Kaggle BCCD dataset across four WBC subtypes, three CNN architectures, four machine learning classifiers, and multiple feature configurations, with rigorous reporting of nine performance metrics.

The key findings of this study are:

- Handcrafted features (LBP + GLCM + Color Histogram + ORB) paired with SVM achieved the highest accuracy of 98.59% and AUC-ROC of 0.9996, demonstrating that carefully engineered domain-specific descriptors remain highly competitive for Giemsa-stained blood cell classification.
- Among CNN architectures, EfficientNetB0 outperformed both VGG16 and ResNet50, achieving 97.64% accuracy with SVM, confirming the superiority of compound-scaled architectures for medical microscopy.
- SVM consistently dominated across all feature configurations, highlighting its suitability for high-dimensional, linearly separable medical imaging feature spaces.
- Five-fold cross-validation confirmed the robustness and generalizability of the proposed framework, with consistent performance across data splits.

Several directions for future work are identified:

- Investigation of Vision Transformer (ViT) and hybrid CNN-Transformer architectures, which have demonstrated state-of-the-art performance on fine-grained medical image classification tasks.
- Exploration of end-to-end fine-tuning of pretrained CNNs on the target dataset, as opposed to fixed-weight feature extraction, to potentially improve deep feature discriminability.
- Extension of the framework to five-class (including Basophil) or pathological cell classification, and validation on larger, multi-institutional datasets to assess clinical translation potential.
- Development of a real-time inference system deployable on edge devices for point-of-care blood analysis in resource-limited clinical settings.

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